

Angles Formed by Parallel Lines and a Transversal

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When two **parallel lines** are cut by a **transversal**, several pairs of angles are formed.

Corresponding Angles

Corresponding angles are equal in measure.

Alternate Interior Angles

Alternate interior angles are equal in measure.

Alternate Exterior Angles

Alternate exterior angles are equal in measure.

Same-Side Interior Angles

Same-side interior angles, also called **consecutive interior angles**, are supplementary.

Same-Side Exterior Angles

Same-side exterior angles, also called **consecutive exterior angles**, are supplementary.

Vertical Angles

Vertical angles are always equal.

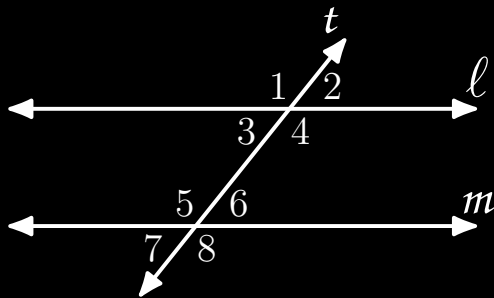
Linear Pairs

Linear pairs of angles are supplementary.

Examples

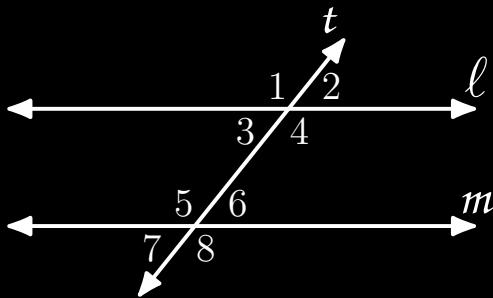
Example 1

Determine the relationship between $\angle 1$ and $\angle 5$.



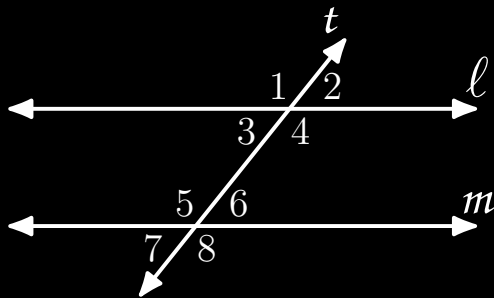
Example 2

Determine the relationship between $\angle 3$ and $\angle 6$.



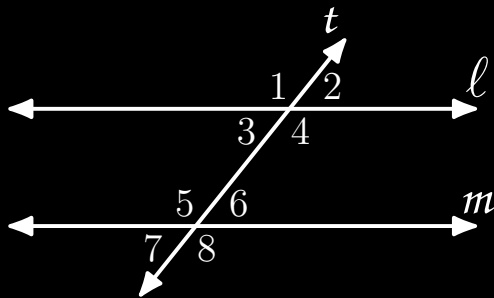
Example 3

Determine the relationship between $\angle 2$ and $\angle 8$.



Example 4

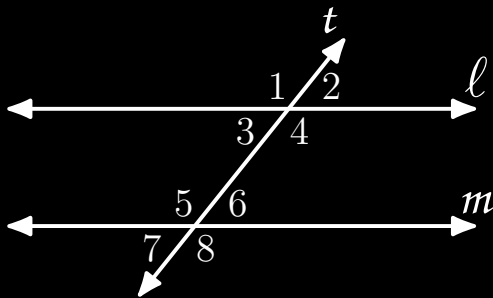
Determine the relationship between $\angle 4$ and $\angle 6$.



Practice Exercises

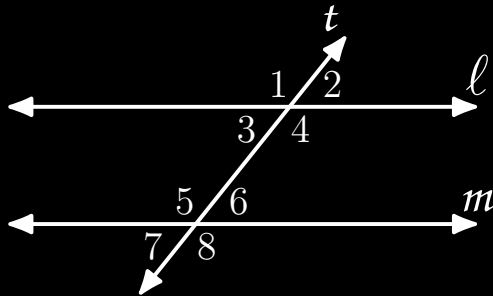
Practice 1

Determine the relationship between $\angle 1$ and $\angle 5$.



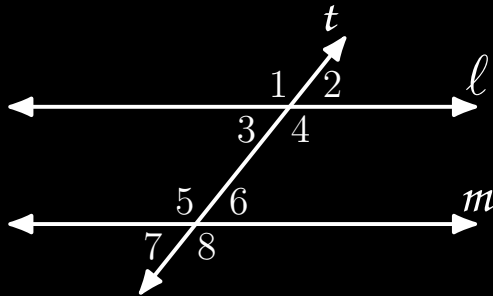
Practice 2

Determine the relationship between $\angle 3$ and $\angle 6$.



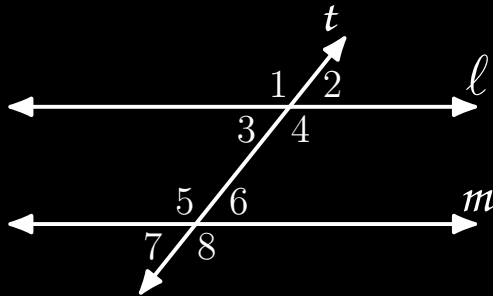
Practice 3

Determine the relationship between $\angle 2$ and $\angle 8$.



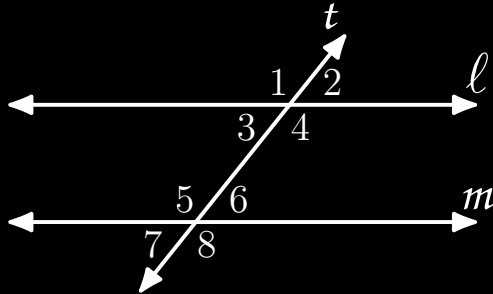
Practice 4

Determine the relationship between $\angle 4$ and $\angle 6$.



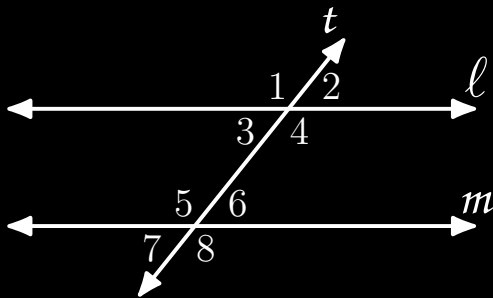
Practice 5

Determine the relationship between $\angle 1$ and $\angle 3$.



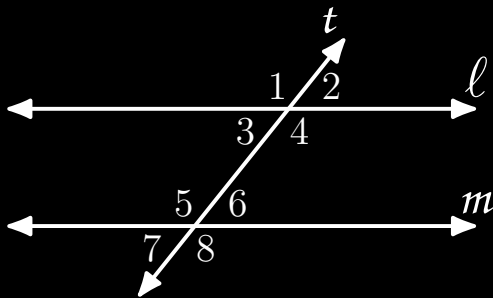
Practice 6

Determine the relationship between $\angle 2$ and $\angle 4$.



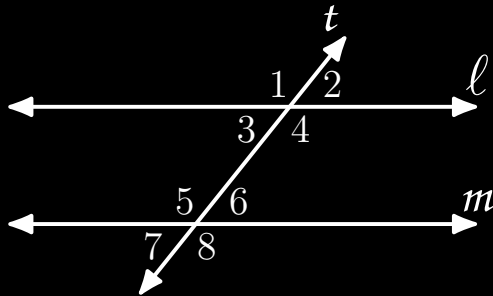
Practice 7

Determine the relationship between $\angle 5$ and $\angle 7$.



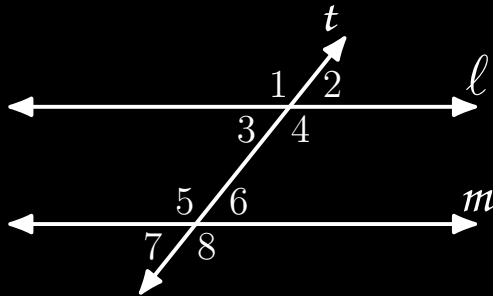
Practice 8

Determine the relationship between $\angle 3$ and $\angle 5$.



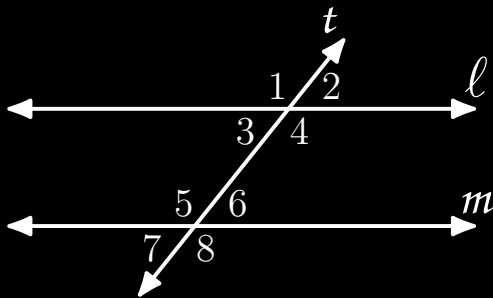
Practice 9

Determine the relationship between $\angle 1$ and $\angle 8$.



Practice 10

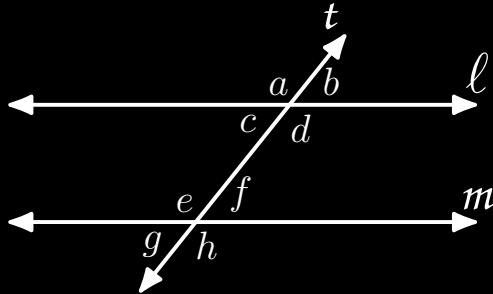
Determine the relationship between $\angle 4$ and $\angle 5$.



Activities

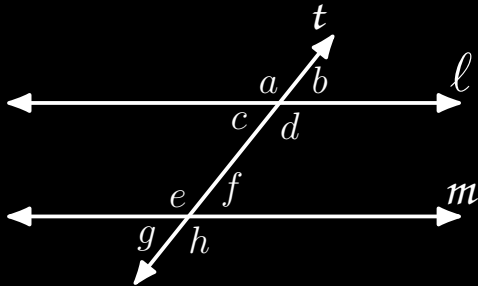
Activity 1

Determine the relationship between $\angle a$ and $\angle e$.



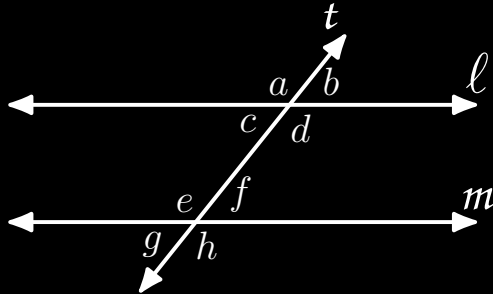
Activity 2

Determine the relationship between $\angle c$ and $\angle f$.



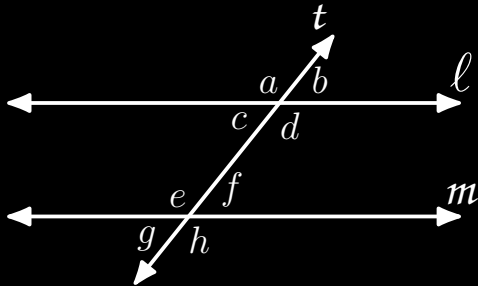
Activity 3

Determine the relationship between $\angle b$ and $\angle h$.



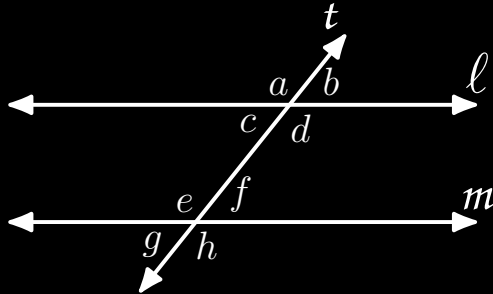
Activity 4

Determine the relationship between $\angle d$ and $\angle f$.



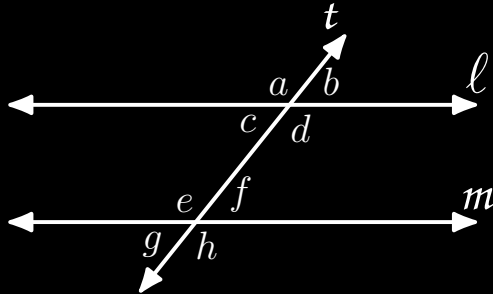
Activity 5

Determine the relationship between $\angle a$ and $\angle c$.



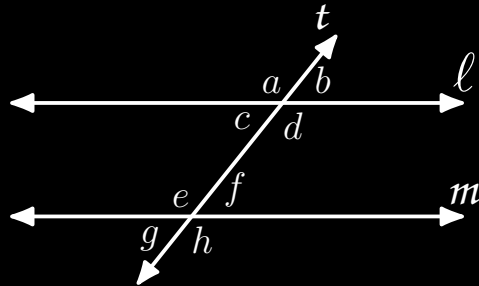
Activity 6

Determine the relationship between $\angle b$ and $\angle d$.



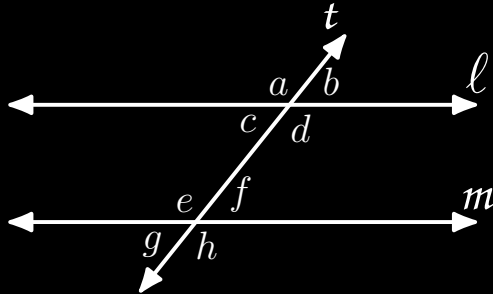
Activity 7

Determine the relationship between $\angle e$ and $\angle g$.



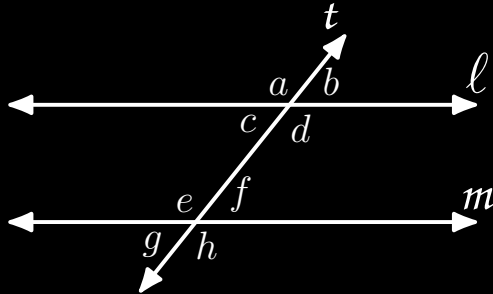
Activity 8

Determine the relationship between $\angle c$ and $\angle e$.



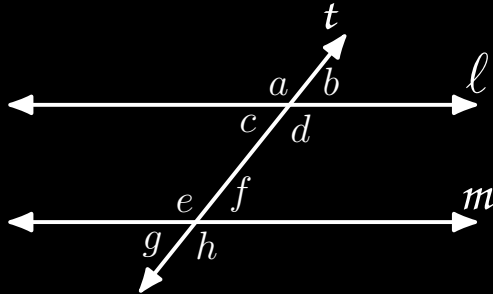
Activity 9

Determine the relationship between $\angle a$ and $\angle h$.



Activity 10

Determine the relationship between $\angle d$ and $\angle e$.



Thank you!